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# Filename.....: ITAI2376_L02_OlisBhari_TeamWork_V1.ipynb
# Language.....: Python
# Tools.....: Visual Studio Code (VSC)
#               : Google Colab
# Class.....: ITAI-2376 Deep Learning in Artificial Intelligence
# Semester.....: Fall 2026
# Class Type....: Online
# Instructor....: Viswanatha Rao
# Team Name.....: Tiger
# Leader.....: Robert Thompson
# Team Member...: Emmanuel Ababio Atta Njeh
# Team Membere..: Olis Bahari
# Version.....: V1.0
# Purpose.....: L02 Exploring Deep Learning Tools: A No-Code Introduction to TensorFlow and Keras
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```
# Import necessary libraries
from tensorflow.keras.applications.vgg16 import VGG16, preprocess_input, decode_predictions
from tensorflow.keras.preprocessing.image import load_img, img_to_array
from IPython.display import display
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image, ImageFilter
import io
import ipywidgets as widgets
```

✓ Introduction

This lab is designed to introduce you to the basics of deep learning by interacting with a pre-built model. You'll understand the workflow of a deep learning project, including data preprocessing, model architecture, and making predictions. The goal is to familiarize yourself with the basics of deep learning without writing any code.

```
# Load the VGG16 model
model = VGG16(weights='imagenet')

# Display the model architecture
model.summary()
```

Model: "vgg16"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0
block1_conv1 (Conv2D)	(None, 224, 224, 64)	1,792
block1_conv2 (Conv2D)	(None, 224, 224, 64)	36,928
block1_pool (MaxPooling2D)	(None, 112, 112, 64)	0
block2_conv1 (Conv2D)	(None, 112, 112, 128)	73,856
block2_conv2 (Conv2D)	(None, 112, 112, 128)	147,584
block2_pool (MaxPooling2D)	(None, 56, 56, 128)	0
block3_conv1 (Conv2D)	(None, 56, 56, 256)	295,168
block3_conv2 (Conv2D)	(None, 56, 56, 256)	590,080
block3_conv3 (Conv2D)	(None, 56, 56, 256)	590,080
block3_pool (MaxPooling2D)	(None, 28, 28, 256)	0
block4_conv1 (Conv2D)	(None, 28, 28, 512)	1,180,160
block4_conv2 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_conv3 (Conv2D)	(None, 28, 28, 512)	2,359,808
block4_pool (MaxPooling2D)	(None, 14, 14, 512)	0
block5_conv1 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv2 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_conv3 (Conv2D)	(None, 14, 14, 512)	2,359,808
block5_pool (MaxPooling2D)	(None, 7, 7, 512)	0
flatten (Flatten)	(None, 25088)	0
fc1 (Dense)	(None, 4096)	102,764,544
fc2 (Dense)	(None, 4096)	16,781,312
predictions (Dense)	(None, 1000)	4,097,000

Total params: 138,357,544 (527.79 MB)

Trainable params: 138,357,544 (527.79 MB)

Non-trainable params: 0 (0.00 B)

```
# Load and preprocess an image
import os

def load_and_preprocess_image(image_path):
    # Load the image
    img = load_img(image_path, target_size=(224, 224))

    # Convert the image to a numpy array
    img_array = img_to_array(img)

    # Expand dimensions to fit the model input
    img_array = np.expand_dims(img_array, axis=0)

    # Preprocess the image
    img_array = preprocess_input(img_array)

    return img, img_array
```

```
# Upload button to load images
upload = widgets.FileUpload(accept='image/*', multiple=False)
display(upload)

# Button to make predictions
predict_button = widgets.Button(description="Make Prediction")
display(predict_button)
```

```

output = widgets.Output()
display(output)

# Function to handle button click
def on_click(change):
    output.clear_output()
    with output:
        if len(upload.value) == 0:
            print("Please upload an image first.")
            return

        # ipywidgets >= 8 stores upload.value as a tuple of dicts.
        # ipywidgets < 8 stores it as a dict keyed by filename.
        if isinstance(upload.value, dict):
            img_data = list(upload.value.values())[0]['content']
        else:
            img_data = upload.value[0]['content']

        img = Image.open(io.BytesIO(img_data)).convert('RGB')
        img = img.resize((224, 224))

        # Preprocess and predict
        img_array = img_to_array(img)
        img_array = np.expand_dims(img_array, axis=0)
        img_array = preprocess_input(img_array)
        predictions = model.predict(img_array)
        decoded_predictions = decode_predictions(predictions, top=3)[0]

        # Display the uploaded image and predictions
        plt.imshow(img)
        plt.axis('off')
        plt.show()
        print(decoded_predictions)

predict_button.on_click(on_click)

```

Upload (1)

Make Prediction

1/1 ————— 0s 881ms/step

1/1 ————— 1s 900ms/step

Downloading data from https://storage.googleapis.com/download.tensorflow.org/data/imagenet_class_index.json

0/35363 ————— 0s 0s/step

35363/35363 ————— 0s 0us/step

